McLeod County as a Cautionary Case: What a Densely Sampled Semi-Rural Community Reveals About the Limits of Genomic Surveillance

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Abstract

Genomic surveillance has been imperative for tracing SARS-CoV-2 transmission and evolution, yet most sequencing efforts have focused on large population centers, leaving a lack of clarity about how rural communities both experience and propagate viral spread. McLeod County, Minnesota is a semi-rural region which achieved an unusually high sampling proportion during the SARS-CoV-2 pandemic, providing a unique opportunity to evaluate whether a small, well-sampled population could serve as a microcosm of wider viral transmission dynamics Using over 1,500 SARS-CoV-2 genomes collected between 2020 and 2024 from McLeod county, we utilized phylogenetic methods to compare variant frequency trends between McLeod and the state, and infer virus introductions into McLeod county to characterize the relationship between sampling depth and inferential confidence. Variant frequency trajectories in McLeod mirrored statewide trends across major Omicron sublineages. However, even with dense viral sampling, we could not confidently infer the true number of introductions into the county, suggesting that introduction counts had not stabilized even at our maximum sampling depth. This finding indicates that even a county with a sampling proportion far exceeding the national average cannot confidently characterize its own transmission dynamics using genomic data alone. Analyzed together, these findings reframe McLeod County not as a validated microcosm but as a productive cautionary case. Even the best semi-rural surveillance scenario we could construct revealed clear boundaries on what genomic data alone can tell us. Future work that grapples honestly with these boundaries and develops analytical frameworks calibrated to realistic sampling constraints will be better positioned to extract reliable inference from the semi-rural communities that genomic epidemiology has historically underserved.

Introduction

The coronavirus disease 2019 (COVID-19), caused by SARS-CoV-2, was first reported in Wuhan, China, in December 2019 (Wu *et al*., 2020). The pandemic unleashed an unprecedented global surge in pathogen genome sequencing in order to accurately generate vast quantities of publicly available SARS-CoV-2 genomic data (Tosta *et al.*, 2023). Large-scale, rapid sequencing efforts directly shaped public health decision-making and policymaking by providing real-time information about viral spread, evolution, and emergence of novel lineages. Phylogenetic and phylodynamic analyses allowed researchers to examine the evolutionary relationships embedded within sampled genomes. When combined with epidemiologic and demographic data, these approaches enabled quantification of viral transmission, identification of outbreak sources, estimation of growth rates and effective reproductive numbers, tracking of variants of concern, and measurement of the impact of interventions such as vaccination (Attwood *et al.*, 2022).

Despite this progress, important gaps in genomic surveillance remained. Many regions, particularly rural or semi-rural areas, had inconsistent sampling, delayed reporting, or incomplete linkage between viral genomes and high-resolution metadata (Ling-Hu *et al.,* 2022). Consequently, highly sampled communities were rare, even though transmission patterns varied substantially across geographic and demographic groups (Tang *et al.,* 2023). This disparity represented a fundamental obstacle to understanding how pathogens move through and between communities of different sizes, densities, and connectivity.

McLeod County, Minnesota, is a small, semi-rural county that belongs to the Minneapolis-St. Paul micropolitan statistical area and represents a rare exception to this pattern of undersampling. McLeod achieved an exceptionally high SARS-CoV-2 genomic sampling proportion (~25-30%), accompanied by detailed metadata including age, vaccination status, and precise geographic location. This level of sampling was uncommon and enabled investigation of hyperlocal transmission while also providing an opportunity to evaluate how sampling intensity influences the reliability of phylogenetic inference.

This study focused on SARS-CoV-2 transmission in McLeod County to determine whether a densely sampled semi-rural community could function as a microcosm of broader regional and national transmission dynamics. We approached this question through four sub-questions: (1) how did sequencing coverage vary across the pandemic timeline; (2) how was transmission in McLeod connected to the rest of Minnesota and the country; (3) how many local sequences are needed before inferred introduction counts stabilize and reliably reflect true transmission dynamics; and (4) did the rise and fall of major variants in McLeod mirror statewide Minnesota frequency trajectories?

To address these questions, we integrated computational tools including Nextstrain, Baltic, and a rarefaction experiment in which McLeod sequences were incrementally added to phylogenetic builds with fixed contextual backgrounds. Rather than confirming that McLeod County behaves as a representative microcosm, our results reveal that in even the best-case semi-rural sampling scenario we could construct, introduction counts never stabilized, and variant frequency comparisons could not be interpreted with confidence. McLeod County thus offers a cautionary lesson about the fundamental limits of semi-rural genomic surveillance and the practical impossibility of achieving the sampling depth that reliable transmission inference would require.

Materials and Methods

Key Resources Table

Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Biological sample (SARS-CoV-2 virus)	12,229 strains from SARS-CoV-2 positive patients in Minnesota and 5,760 strains from SARS-CoV-2 positive patients from other US states	Minnesota Department of Health	Sequences were pulled from public repositories in Genbank and GISAID	Full metadata for each sequence is available in this github repository
Biological sample (SARS-CoV-2 metadata)	Demographic information about 12,229 strains from SARS-CoV-2 positive patients in Minnesota and 5,760 strains from SARS-CoV-2 positive patients from other US states	Minnesota Department of Health		Full metadata for each sequence is available in this github repository

Biological sample (SARS-CoV-2 virus)	Publicly available SARS-CoV-2 genomes	Wu *et al.,* 2020	Sequences were pulled from public repositories in Genbank and GISAID under accession MN908947	
Software, algorithm	MAFFT	Katoh *et al.,* 2002	https://mafft.cbrc.jp/alignment/software/	
Software, algorithm	TreeTime	Sagulenko *et al.,* 2018	https://github.com/neherlab/treetime	
Software, algorithm	IQTREE	Nguyen *et al.,* 2015	https://iqtree.github.io/	
Software, algorithm	Baltic	https://github.com/evogytis/baltic	Python Library	
Software, algorithm	Github repo with scripts used to analyze data and generate figures for this manuscript	This paper	https://github.com/moncla-lab/WI-SARS-CoV-2-Phylodynamics/tree/main/Mcleod	This github repository contains all of the code used to generate figures and perform the analyses described in this paper

Data and code availability

All of the code used for preprocessing metadata, generating Nextstrain builds, and parsing phylogenetic transitions was written in Python (v3.10) and Nextstrain Augur/Auspice (v23). Analysis pipelines, including custom scripts for Baltic parsing and metadata harmonization, can be found at https://github.com/moncla-lab/WI-SARS-CoV-2-Phylodynamics/tree/main/Mcleod

Biological sample data including positive patient sampling in Minnesota came from the Minnesota Department of Health under a data use agreement. Associated information includes date (year-month-day), geographic location at the ZIP-code or county level, gender, age, vaccination count, and urban/rural status for all Minnesota samples. Due to the agreement, specific demographic information is unavailable in the public repository. In the complete data set, 1,616 strains originated in McLeod County, MN, 10,613 strains originated in remaining counties excluding McLeod County (greater-MN), and 5,760 strains originated in other U.S. states (non-MN).

Comparison of Case Counts and Sequencing Coverage

In order to best understand the sampling proportion and sequence coverage of Mcleod County, MN, daily confirmed COVID-19 case counts were obtained from the Minnesota Department of Health COVID-19 surveillance reports. These case counts were binned based on MMWR epidemiologic week and matched by date to the number of SARS-CoV-2 genomic sequences available from the same interval. The difference between the case count and the sequence count was calculated in a custom python script. Visualization of this difference was performed in Python (v3.10) using pandas and Plotly, where interactive graphs were also generated. Bar charts representing weekly sequences were overlaid with line plots showing weekly case counts. A secondary subplot displayed the calculated difference (cases - sequences). Custom hover text was used to display week-specific counts, differences, and metadata. The resulting figure was exported as an interactive HTML dashboard, allowing detailed review of sequencing intensity and sampling proportion relative to the infection burden over the course of the pandemic (Figure 1). Another plot was generated to show the proportion of cases sequenced per week, color-coded by coverage tier: low (<10%), moderate (10-30%), and high (>30%). The resulting figure was also exported as an interactive HTML dashboard (Figure 2).

Phylogenetic Reconstruction Using Nextstrian and Transition Parsing with Baltic

Phylogenetic trees were generated using the Nextstrain Augur pipeline. All McLeod County sequences were included in each build, supplemented with contextual sequences from greater Minnesota (greater-MN), Twin Cities metropolitan counties (TC_county), and other U.S. states (non-MN), downsampled by time and region to avoid overrepresentation while maintaining phylogenetic breadth. Trees were generated using MAFFT for alignment, IQ-TREE for maximum likelihood phylogeny, and TreeTime for time-resolved reconstruction. Metadata files were harmonized to allow trait mapping for vaccination status, RUCA code, state, urban/rural classification, age group, SARS-CoV-2 strain, pango lineage, and source region. Discrete trait analysis (DTA) was performed on multiple traits including Twin City status, urban/rural classification, and county status, assigning sequences to geographic and demographic categories to enable ancestral state reconstruction. For Twin City status specifically, sequences were assigned to one of four categories: McLeod, TC_county, greater-MN, or non-MN, allowing directional viral transitions into and out of McLeod to be inferred across the tree. Auspice JSON outputs were used to color nodes according to these traits to visualize cluster structure and transitions into and out of McLeod.

Temporal lineage frequencies were estimated using augur frequencies, which applies a time-windowed smoothing model to infer the relative abundance of SARS-CoV-2 lineages through time. Frequency models were visualized in Auspice and as line plots with clade frequency on the y-axis and time on the x-axis, computed separately for McLeod, greater-MN, TC_county, and non-MN samples to enable direct trajectory comparison across regions.

Directional viral movement into and out of McLeod County was quantified using the Baltic phylogenetic library. Baltic scripts extracted ancestral state assignments and node-to-node transitions between geographic categories for each Nextstrain time-scaled tree.

Rarefaction Experiment: Sampling Sufficiency for Reliable Inference

To determine how many local sequences from McLeod County are needed before inferred introduction counts stabilize and reliably reflect true transmission dynamics, we conducted a rarefaction experiment specifically from March 20, 2022 to April 16, 2023, a section of the Omicron period, when sampling coverage in McLeod was highest.

A large contextual dataset was used as the backbone for the rarefaction analysis which contained 2000 sequences from greater-MN, 2000 sequences from Twin Cities, and 2000 sequences from non-MN. McLeod County sequences were then subsampled in incremental steps of 50 sequences, from a minimum of 50 up to a maximum of 700 sequences. At each increment, five independent iterations were generated to account for variation in sequence selection, yielding a total of 70 phylogenetic builds (14 increments × 5 iterations). For each build, a time-resolved phylogenetic tree was reconstructed, and Baltic was used to parse directional viral transitions and count the number of inferred introductions into McLeod County from each of the three source regions (TC_county, greater-MN, and non-MN). The number of introductions was plotted on the y-axis against the number of McLeod sequences included on the x-axis, producing rarefaction curves. If introduction counts were to stabilize, the curve would be expected to plateau at an asymptote, identifying a minimum sampling threshold sufficient for reliable inference. Exports from McLeod into each destination region were also inferred and plotted using the same framework.

Results

Comparison of Case Counts and Sequencing Coverage

To evaluate sampling proportion over time, we compared weekly SARS-CoV-2 genomic sequence counts to confirmed case counts across the full pandemic timeline (Figure 1A). Raw counts showed that case numbers far exceeded sequence counts during the Alpha (September 2020 to September 2021) and Delta (June 2021 to November 2021) waves, with some weeks in the beginning of the Omicron (November 2021 to March 2022) showing case-to-sequence ratios that made phylogenetic inference unreliable for those time periods with the lowest sampling proportions. Sequencing coverage was substantially higher during the middle and end of the Omicron wave, with multiple weeks reaching moderate-to-high coverage tiers and the overall sampling proportion across the pandemic averaging approximately 25–30% (Figure 1B). This improvement in coverage during Omicron (March 2022 to April 2023) informed our decision to focus downstream analyses on this period. The coverage plots revealed notable dips during high-case-count surges, indicating that the practical challenge of keeping pace with sequencing demand during peak transmission was not fully overcome even in this exceptionally well-sampled county. This level of genomic coverage was substantially higher than typical U.S. county-level sampling (Lipsitch *et al.*, 2024).

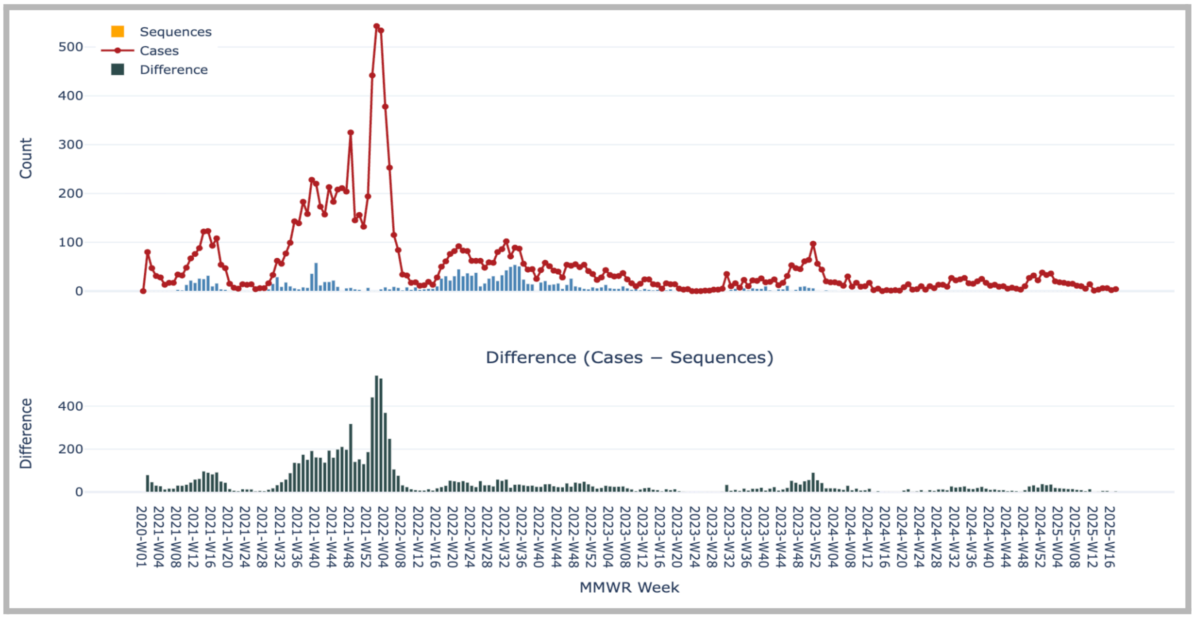

Figure 1a.

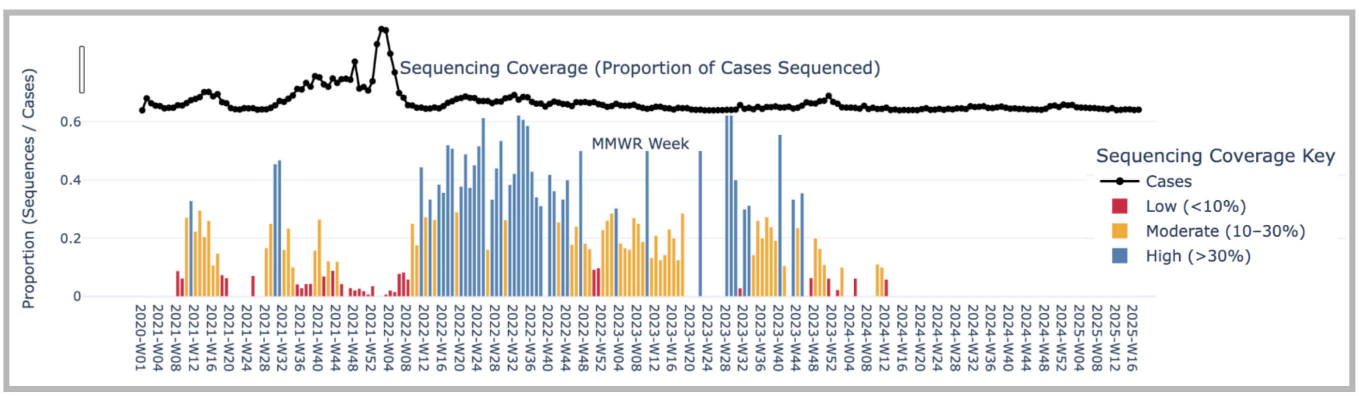

Figure 1b.

Figure 1: Sequences vs Cases per MMWR Week and the Corresponding Difference (with link to interactive version). Sequencing Coverage Proportion by MMWR Week.

(A) Weekly SARS-CoV-2 genomic sequence counts compared to confirmed case counts in McLeod County, MN across the full pandemic timeline. The bottom panel shows the raw difference between cases and sequences per MMWR epidemiologic week, highlighting periods of low proportional coverage, particularly during high-transmission waves in 2021. (B) Proportion of confirmed COVID-19 cases genomically sequenced per MMWR week in McLeod County, MN. Coverage is categorized as low (<10%), moderate (10–30%), or high (>30%). The Omicron period showed the most sustained moderate-to-high coverage and was therefore selected as the focus of the rarefaction analysis.

Phylogenetic Reconstruction Reveals Importation and Exportation Events

Constructing phylogenies with McLeod County sequence data and contextual sequences from other counties and states allowed us to visualize transmission dynamics between geographic categories (Figure 2). McLeod County samples (orange) clustered frequently alongside samples from the Twin Cities (TC_county, gold), greater-MN (light purple), and non-MN states (dark purple) throughout the pandemic timeline, indicating that McLeod was not epidemiologically isolated but rather deeply embedded in regional and national transmission networks. The interleaving of McLeod samples with non-MN samples across the tree suggested recurring introductions from outside Minnesota, a pattern consistent with the movement of people between urban population centers and surrounding communities.

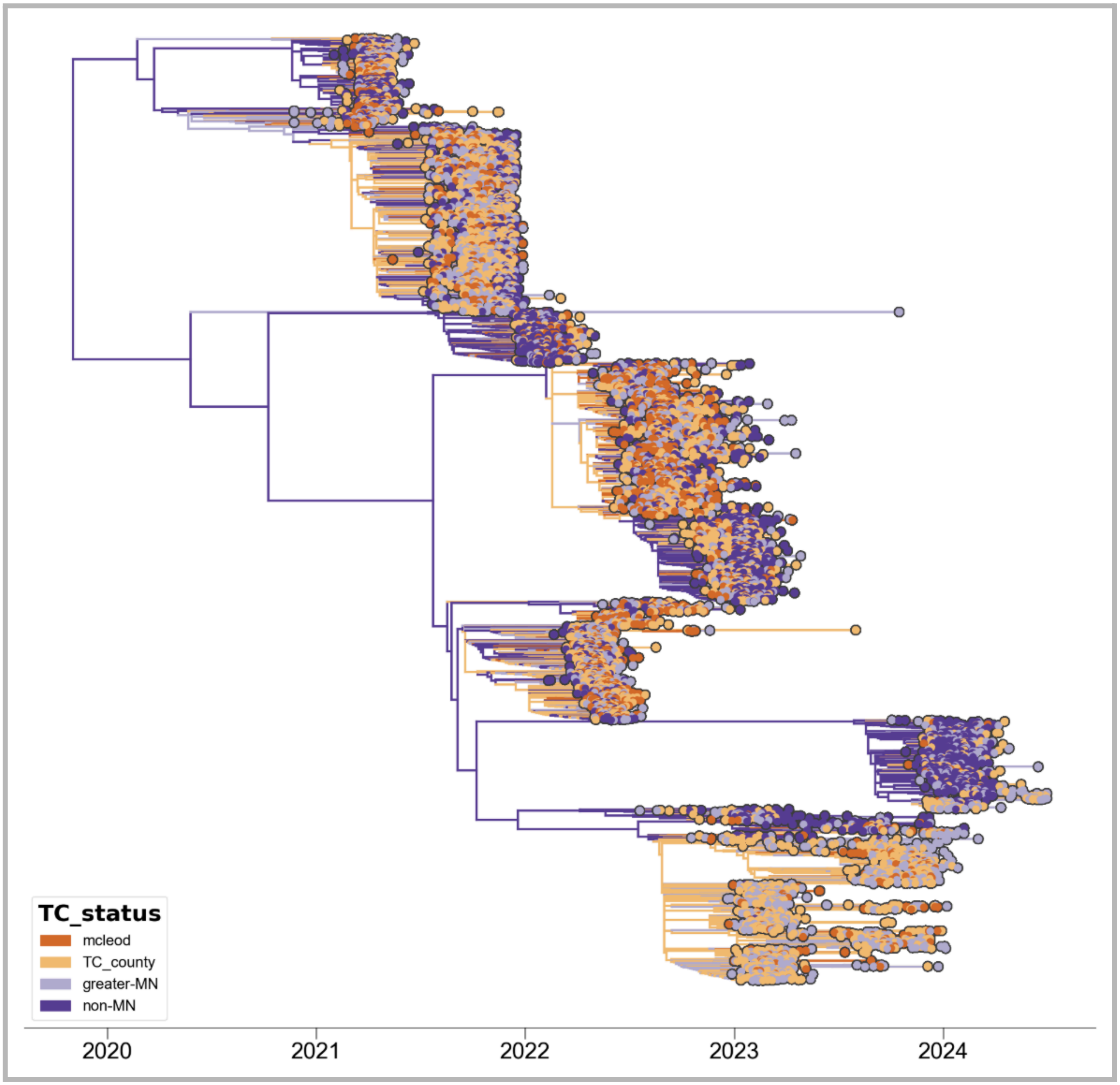

Figure 2: Time-resolved phylogenetic tree colored by Twin City status (mcleod = orange, TC_county = gold, greater-MN = light purple, non-MN = dark purple). X-axis: date (2020–2024).

Time-resolved maximum likelihood phylogeny of 7,956 SARS-CoV-2 genomes sampled between November 2020 and June 2024, colored by source region. McLeod County samples (orange) are distributed throughout the tree and cluster frequently with Twin Cities (gold), greater-MN (light purple), and non-MN (dark purple) samples, indicating repeated introductions from external sources and deep connectivity within regional transmission networks.

Rarefaction Curves Reveal That Introduction Counts Did Not Stabilize Even at Maximum Sampling Depth

To determine how many local sequences are needed before inferred introduction counts reliably reflect true transmission dynamics, we conducted a rarefaction experiment during the Omicron period, incrementally increasing McLeod sequences from 50 to 700 in steps of 50, with five iterations at each level. We anticipated that introduction counts from each source region would plateau at an asymptote, identifying a minimum sufficient sampling threshold. Instead, rarefaction curves for all three source regions, Twin Cities counties, greater-MN, and non-MN, continued to rise throughout the full range of McLeod sequences included, showing no clear asymptote even at 700 sequences (Figure 3).

Introductions inferred from non-MN and greater-MN were the most numerous, while introductions from Twin Cities counties were consistently lower across all sampling levels. While the numbers of introductions do continue to increase through the entire range, we do see the ratios of introduction source emerge early and hold throughout most of the experiment, with similar levels of introductions from greater MN and non-MN and lower rates of introduction from the Twin Cities.

Exports from McLeod to all three destination regions were substantially fewer than introductions at every sampling increment, suggesting that McLeod was primarily a recipient rather than a source of viral spread, though export curves also continued to rise without stabilizing (Figure 3). Similarly, we see consistent proportions of exportations across the rarefaction, with consistently the highest rates into greater-MN and lower and even rates of exportation to Twin Cities and non-MN. Error bars from the five iterations at each increment were relatively narrow, indicating that this pattern was not driven by random variation in sequence selection but rather reflects a genuine property of the data that more sequences consistently revealed more inferred introductions, with little sign of saturation.

The absence of an asymptote indicates that at 700 sequences, already a substantial number for a county of McLeod's size, introduction counts had not yet converged on a stable estimate. Extrapolating the curve suggests that a true asymptote, if one exists, would require a number of local sequences that approaches the total number of cases in the county during the Omicron period. In other words, reliable inference about transmission dynamics would likely require a sampling proportion far beyond what McLeod achieved, and potentially beyond what is practically achievable in any surveillance setting.

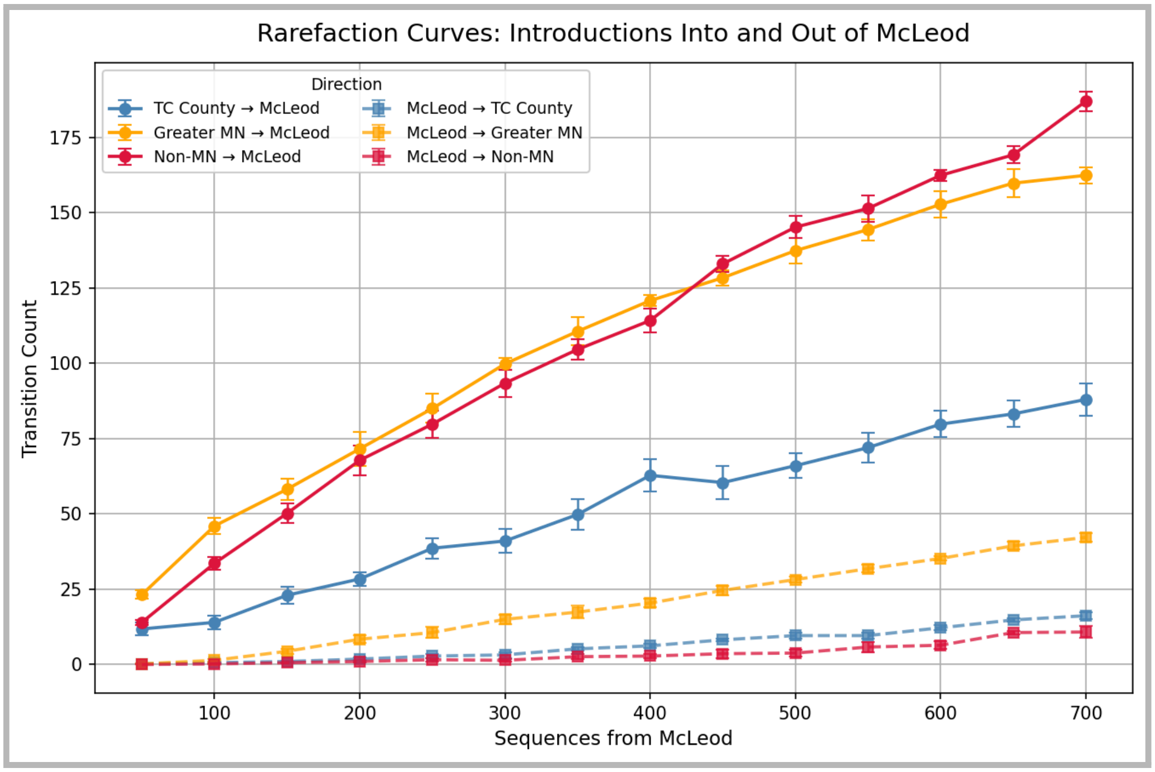

Figure 3: Combined rarefaction curves for all transition directions (introductions and exports) on one plot, using solid lines for introductions and dashed lines for exports, colored by region.

Combined rarefaction curves showing all six directional transition categories (introductions into McLeod and exports from McLeod, for each of the three source/destination regions). Introductions (solid lines) substantially outnumber exports (dashed lines) across all sampling levels. Neither introductions nor exports reach an asymptote through 700 McLeod sequences.

Variant Frequency Trajectories in McLeod Visually Track Statewide Trends, But Cannot Be Interpreted with Confidence

Temporal lineage frequency plots were generated to compare the rise and fall of major SARS-CoV-2 variants in McLeod County against statewide Minnesota trends. Visually, the trajectories of Alpha (GRY), Delta (GK), and Omicron (GRA) in McLeod closely tracked those seen across all greater-MN, TC_county, and non-MN samples (Figure 4). When variant-specific frequency curves were compared across location categories using the same phylogenetic build, the clade trajectories showed broadly similar timing, with McLeod's Omicron 50% frequency threshold appearing to precede that of non-MN samples by approximately 14 days and follow that of TC_county and greater-MN samples by approximately 21–28 days (Figure 5).

However, these observations must be interpreted with significant caution. The contextual sequences included in this build, from greater-MN, TC_county, and non-MN, were themselves subsampled and do not represent complete or proportional coverage of those regions. The apparent time lags between regions may therefore reflect sampling artifacts rather than true epidemiological differences in the timing of variant emergence. Because we cannot verify that the contextual data captures the full variant frequency trajectory of those regions, we cannot confidently attribute the observed differences in crossing times to biological or behavioral factors. The visual similarity in trajectory shapes between McLeod and statewide trends is noted as an observation of interest, but no firm conclusions about the timing or direction of variant spread can be drawn from this analysis.

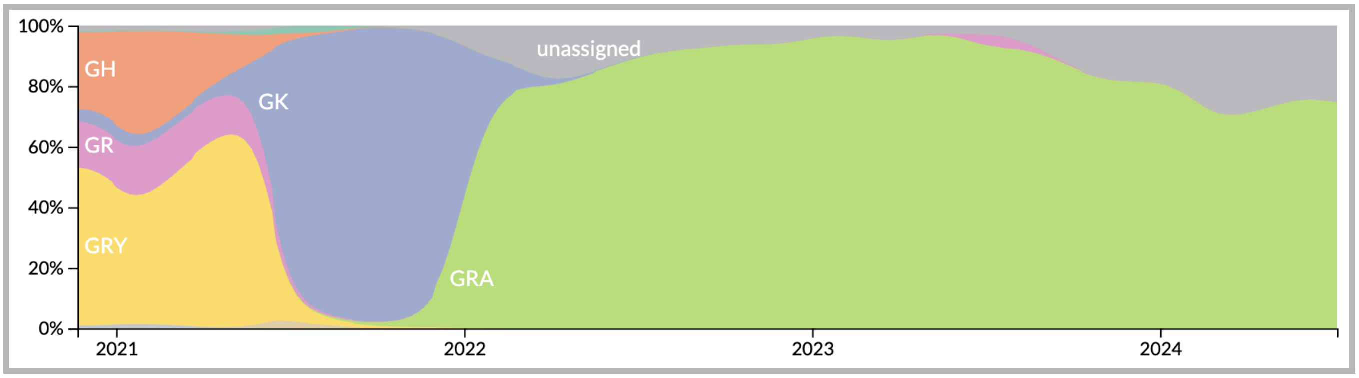

Figure 4a.

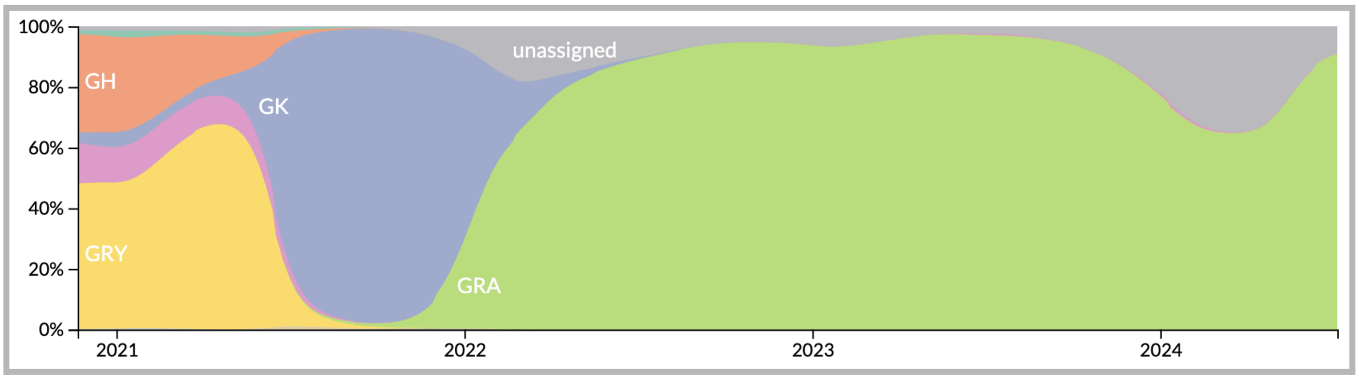

Figure 4b.

Figure 4: McLeod County only and Minnesota only colored frequencies by SARS-CoV-2 Strain and normalized to 100% at each time point for 5958 out of a total of 7956 tips (GRA represents Omicron, GK represents Delta, and GRY represents Alpha).

(A) Temporal SARS-CoV-2 clade frequency plots for McLeod County, normalized to 100% at each time point. (B) Temporal SARS-CoV-2 clade frequency plots for all of Minnesota), normalized to 100% at each time point. The rise of Alpha (GRY), Delta (GK), and Omicron (GRA) followed broadly similar trajectories in McLeod and statewide, but these visual similarities should not be interpreted as evidence of true epidemiological synchrony given the limitations of contextual sampling.

Figure 5a.

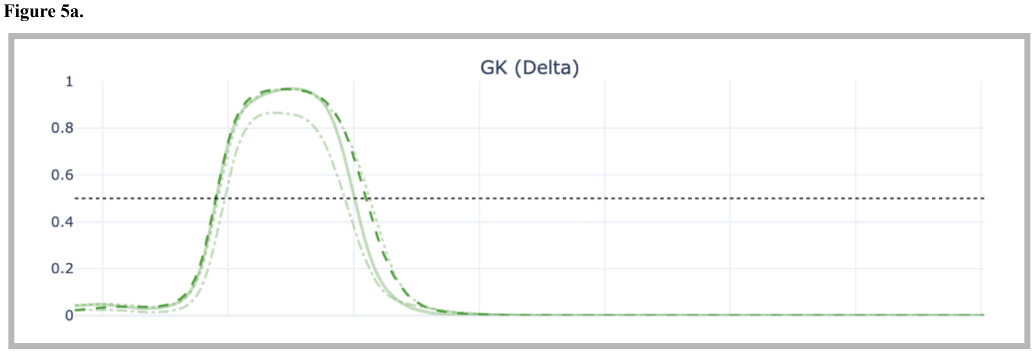

Figure 5b.

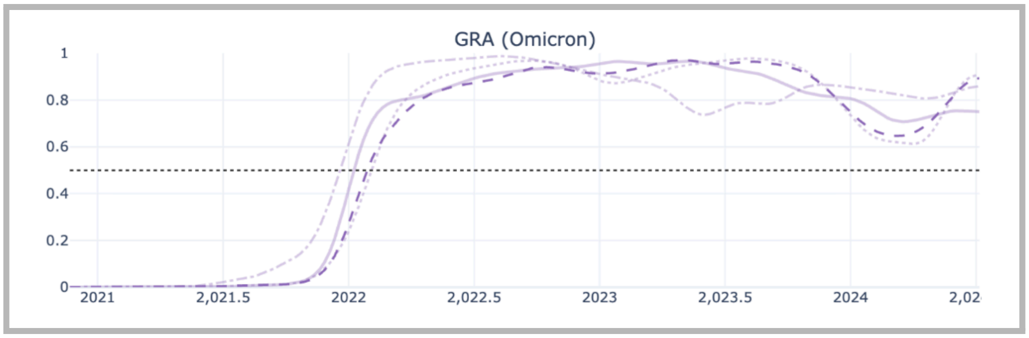

Figure 5c.

Figure 5: Multi-panel line plot showing clade-specific frequency curves by location (McLeod, greaterMN, TC_data, nonMN) for Alpha, Delta, and Omicron separately.

Variant-specific frequency trajectories for Alpha (GRY) (5a.), Delta (GK) (5b.), and Omicron (GRA) (5c.) across four location categories: McLeod County, greater-MN, TC_county, and non-MN. Non-MN crossed approximately 14 days after McLeod, while TC_county and greater-MN crossed approximately 28 and 21 days before McLeod, respectively for the Omicron specific variant (Data not shown). These values are presented descriptively and should not be interpreted as causal given the limitations of contextual sampling.

Discussion

The unusually dense genomic surveillance of McLeod County, Minnesota offered a rare opportunity to ask whether intensive local genomic sampling could make a semi-rural community a reliable microcosm of broader regional transmission dynamics. The answer that emerged from this study is more complex, and more cautionary, than the question implies. Across every analysis we conducted, the data were never quite sufficient to draw firm conclusions, even in the best-case semi-rural surveillance scenario we could construct.

The sequencing coverage analysis established that McLeod's ~25-30% average sampling proportion, while exceptional by national semi-rural standards, was not evenly distributed across time. Coverage dropped during periods of high case counts, precisely when transmission inference would have been most valuable. The Omicron wave represented the most favorable sampling window, which is why we focused our rarefaction experiment there, but even still, the results revealed limits on what we can infer from the data.

The rarefaction experiment was the most direct test of sampling sufficiency, and its results reframe the study's central question. We anticipated that incrementally increasing McLeod sequences in a fixed, fully contextualized build would produce rarefaction curves that plateau, identifying the minimum sampling depth at which introduction counts stabilize and can be trusted as reflective of true transmission dynamics. We supposed that at some point, adding more samples would reveal diminishing marginal information, and the curve would level off. However, that asymptote never appeared. Through 700 McLeod sequences, already a number that most semi-rural counties could not approach, the curves for introductions from all three source regions continued to rise approximately linearly. This means that each new McLeod sequence added to the build continued to reveal previously undetected introduction events, with no sign of saturation.

The implications of this finding suggest that the true number of introductions into McLeod County during the Omicron period was substantially higher than any of our builds captured, and that achieving a stable estimate would require sampling proportions that approach or exceed what is practically achievable in any real-world surveillance context. McLeod County, with its exceptionally strong public health infrastructure and genomic surveillance resources, represents a ceiling for what semi-rural communities can realistically accomplish. If even this county cannot generate the data density needed for stable phylogenetic inference, that ceiling is very low.

The pattern of inferred introductions, while not stabilized, was nonetheless directionally consistent across all sampling levels. Non-MN and greater-MN were the dominant sources of introductions into McLeod at every point on the rarefaction curve, and exports from McLeod to all regions were fewer than introductions. This asymmetry suggests that McLeod functioned primarily as a recipient of viral lineages originating in more densely connected regions, consistent with established patterns of pathogen flow from urban to semi-rural communities (Moreno *et al.,* 2020). However, given that neither the introduction nor the export curves stabilized, the precise magnitudes of these flows remain uncertain, and the relative contributions of each source region may shift further as sampling increases.

The variant frequency analysis offers a visually compelling but epidemiologically ambiguous result. The rise and fall of Alpha, Delta, and Omicron in McLeod closely tracked the patterns seen statewide, and quantitative analysis of Omicron 50% frequency crossing times suggested that McLeod may have preceded non-MN adoption of Omicron by approximately two weeks while lagging behind TC_county and greater-MN by three to four weeks. This ordering is biologically plausible because it would be consistent with a model in which Omicron seeded first in urban cores and spread to semi-rural connected counties like McLeod. However, these interpretations cannot be supported with confidence. The contextual sequences from each region were themselves subsampled and do not reflect the true variant frequency trajectory of those populations. The crossing time differences we observed may be real, or they may reflect gaps in our contextual data. Without complete or representatively proportional sampling of each comparison region, variant timing comparisons are inherently fragile. We present these observations as hypotheses worth revisiting with stronger data, not as conclusions.

These limitations collectively point to a broader challenge for semi-rural genomic epidemiology. The field has made tremendous progress in using genomic data to understand pandemic dynamics, but that progress has been concentrated in settings where dense sampling is feasible, such as large urban centers, national surveillance programs in high-income areas, and well-resourced academic medical centers. The McLeod County study was designed to ask whether semi-rural communities could participate meaningfully in this scientific enterprise, and they still can, because the data are informative, the patterns are suggestive, and the infrastructure is robust. Still, the rarefaction results make clear that "informative" and "sufficient for reliable inference" are not the same.

Future studies should pursue several different directions. First, applying the rarefaction framework developed here to other Minnesota counties with varying sampling proportions would test whether the absence of an asymptote is a general feature of semi-rural phylogenetic inference or specific to McLeod's demographic and geographic context. Second, integrating demographic metadata into cluster-level analyses could help identify which subpopulations are disproportionately involved in introduction events, as the phylogenetic trees colored by these traits suggest patterns worth investigating systematically. Finally, methodological work is needed to develop analytical approaches that can produce reliable transmission inference under realistic sampling constraints. These approaches would explicitly account for incomplete sampling rather than assuming that observed transitions approximate true ones.

This study advances genomic epidemiology by demonstrating, with unusual clarity, where the limits of semi-rural surveillance lie. The value of McLeod County's extraordinary sampling effort is not diminished by the finding that it was still insufficient, rather, that finding is itself the contribution. It establishes a benchmark that if ~25–30% sampling in a small, well-resourced, metadata-rich county cannot produce stable transmission inference, then the field must either develop methods that work with less, or advocate seriously for the resources needed to do more.

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